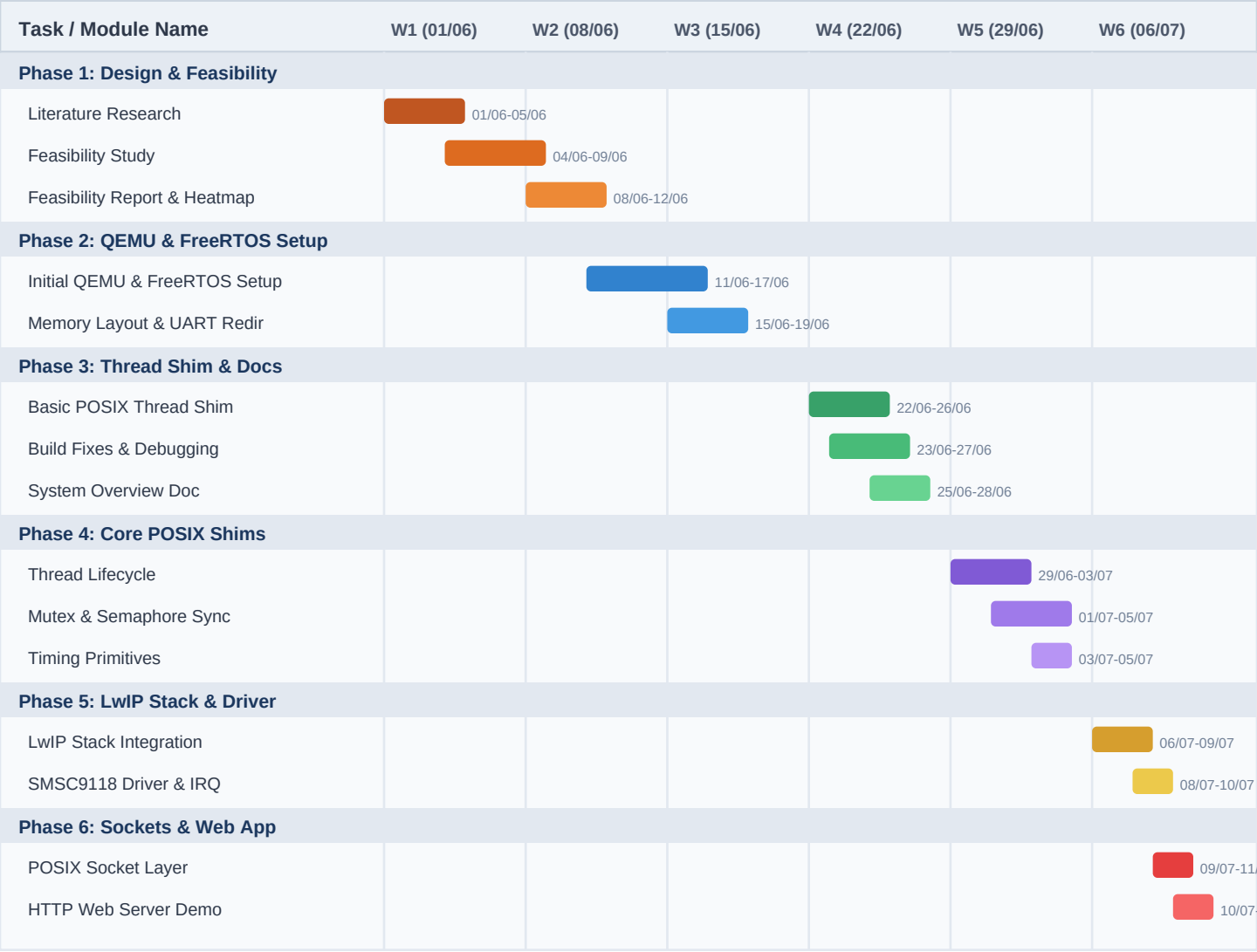


STM32 Simulated Linux: 6-Week Gantt Chart & Project Roadmap

Master schedule detailing the POSIX compatibility layer development on FreeRTOS in QEMU (01-06-2026 – 12-07-2026).

1. Gantt Chart Timeline Diagram



2. Master Schedule Breakdown

ID	Phase	Full Task Name	Key Output / Deliverable
t0	Phase 1	Literature Research	Survey of POSIX RTOS shimming & Cortex-M memory bounds
t1	Phase 1	Research & Feasibility Study	Cortex-M3 RAM budget & POSIX evaluation
t2	Phase 1	Feasibility Report & Heatmap	Feasibility report & heatmap creation
t3	Phase 2	Initial QEMU & FreeRTOS Setup	FreeRTOS Kernel & QEMU MPS2 target setup
t4	Phase 2	Memory Layout & UART Redir	Linker script mps2_m3.ld & main.c UART init
t5	Phase 3	Basic POSIX Thread Translation Shim	pthread_create mapping to xTaskCreate
t6	Phase 3	Build Fixes & Debugging	Toolchain cross-compilation fix
t7	Phase 3	System Overview & Architecture	Architectural guide SYSTEM_OVERVIEW.md

ID	Phase	Full Task Name	Key Output / Deliverable
t8	Phase 4	Thread Lifecycle (join, exit, detach)	Thread registry & lifecycle in main_blinky.c
t9	Phase 4	Mutex & Semaphore Synchronization	pthread_mutex_t & counting sem_t
t10	Phase 4	Timing Primitives (sleep, usleep)	Delay mapping to vTaskDelay ticks
t11	Phase 5	LwIP TCP/IP Stack Integration	LwIP OS layer adaptation in sys_arch.c
t12	Phase 5	SMSC9118 Driver & NVIC IRQ	Ethernet driver in ethernetif.c
t13	Phase 6	POSIX Socket Shim Wrapper Layers	BSD Socket APIs (socket, bind, listen)
t14	Phase 6	HTTP Web Server Demo App	Simulated POSIX Web Server on port 80/8080

3. Detailed Week-by-Week Breakdown

Phase 1: Design, Heatmap & Feasibility Report (01-06 to 12-06-2026)

- Performed literature research on POSIX API shimming over real-time operating systems.
- Evaluated memory constraints and RAM budget (~70-80 KB) of the target board.
- Mapped API conversion strategies, priority scaling, and stack depth allocations for simulating Linux threads.
- Created methodology evaluation heatmap comparing POSIX Shimming vs. Manual Rewriting and Full Emulation.
- Published initial project guidelines and repository setup documentation in README.md.

Phase 2: QEMU Setup & FreeRTOS Installation (11-06 to 19-06-2026)

- Imported the FreeRTOS real-time kernel and template configuration for Cortex-M3 MPS2 AN385 platform.
- Configured linker script mps2_m3.ld defining memory layouts for FLASH (4096K) and SRAM (8192K).
- Overrode stdout bindings in main.c to pipe console prints directly to QEMU UART0 registers.

Phase 3: POSIX Thread Shim, Build Fixes & System Overview (22-06 to 28-06-2026)

- Designed lightweight pthread_create translation mapping POSIX requests directly to FreeRTOS xTaskCreate.
- Fixed type-casting constraints inside pthread_create and updated Makefile compiler flags.
- Created SYSTEM_OVERVIEW.md detailing architectural layout and execution pathways.

Phase 4: Core POSIX Shim Layers (29-06 to 05-07-2026)

- Implemented thread lifecycle APIs in main_blinky.c (pthread_join, exit, detach, self).
- Mapped pthread_mutex_t to FreeRTOS mutexes and built custom sem_t counting semaphores.
- Integrated timing delay wrappers (sleep, usleep) mapped to FreeRTOS scheduler ticks.

Phase 5: LwIP Stack & Driver (06-07 to 10-07-2026)

- Integrated LwIP TCP/IP stack with custom memory configuration tuned in lwipopts.h.
- Built sys_arch.c OS adaptation layer and SMSC9118 Ethernet driver in ethernetif.c.
- Configured interrupt service routines (NVIC IRQ 13) to process incoming packet queues asynchronously.

Phase 6: Sockets & Web Application (09-07 to 12-07-2026)

- Mapped standard Linux BSD sockets (socket, bind, listen, accept, read, write, close) to LwIP.
- Developed a simulated POSIX HTTP Web Server daemon inside main_blinky.c listening on port 80/8080.